

Non-Minimum-Phase Channel DoA Estimation via UCA Phase-Mode Smoothing and Enhanced Subspace Separation

A thesis submitted in partial fulfillment of the requirements for
the award of the degree of

B.Tech.

in

Electronics and Communication Engineering

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MAY 2026

Dedicated to

*my mother, Pournima,
my sister, Vishakha,
and my beloved friend, Chhavi,
who are the pillars of my life.*

CERTIFICATE OF COLLABORATION

This is to certify that the project titled **Non-Minimum-Phase Channel DoA Estimation via UCA Phase-Mode Smoothing and Enhanced Subspace Separation** is a record of the work carried out by

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in collaboration with the **Centre of Excellence in Wireless Technology (CEWiT)**, **Indian Institute of Technology Madras**, and with technical support from **Data Patterns (India) Pvt. Ltd., Chennai**, during the academic year 2025–2026, under the supervision of **Prof. K. Giridhar**.

The work was carried out with technical mentorship from **Ramanathan C.** and **Venkatesh C. R.**, contributing towards the development and analysis of advanced signal processing techniques for line-of-sight (LOS) detection in non-minimum phase channels using uniform circular arrays.

Chinmay Deo is a bona fide student of the **National Institute of Technology, Jalandhar**, pursuing his Bachelor of Technology in Electronics and Communication Engineering. His work demonstrates strong proficiency in array signal processing, statistical estimation, and practical system implementation, with relevance to real-world sensing and radar systems.

Prof. K. Giridhar

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ABSTRACT

Direction-of-arrival (DoA) estimation using compact antenna arrays is severely challenged in non-minimum phase channels, where coherent multipath and interference lead to rank-deficient covariance matrices and failure of conventional eigenstructure-based methods such as MUSIC. This limitation is particularly critical in practical sensing and radar systems operating under strong multipath, fading, and adversarial interference.

This work investigates reliable line-of-sight (LOS) detection using an undersampled five-element Uniform Circular Array (UCA). A unified framework combining phase-mode excitation, mode-space spatial smoothing, forward-backward averaging, and real beamspace MUSIC is proposed to restore the signal subspace under coherence. A dynamic noise-variance-based thresholding scheme is introduced to enable robust signal and noise subspace separation.

The proposed method is evaluated across diverse propagation environments, including specular-dominant and fully scattered channel conditions. Simulation results demonstrate reliable LOS DoA estimation despite strong source coherence and interference. Through mode-space spatial smoothing, forward-backward averaging, and the proposed dynamic thresholding scheme, the framework enables stable subspace separation and improved localization performance under conditions where conventional approaches fail.

The estimator approaches the deterministic Cramér-Rao Bound (CRB) at operational signal-to-noise ratios above 5 dB and maintains appreciable LOS isolation under moderate coherent interference. These results establish a narrow yet practical operating regime for compact sensing systems deployed in challenging propagation environments.

Overall, this work presents a computationally efficient and experimentally validated framework for LOS detection in challenging propagation environments, with applications in passive radar, wireless sensing, and autonomous systems.

Keywords: Direction-of-Arrival Estimation, Uniform Circular Arrays, Phase-Mode Excitation, Mode-Space Spatial Smoothing, Real Beamspace MUSIC, Coherent Sources, Non-Minimum Phase Channels, Passive Radar

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Prof. K. Giridhar**, Department of Electrical Engineering, Indian Institute of Technology Madras, for his invaluable guidance, support, and encouragement throughout the course of this project. His insights into array signal processing and system-level design were instrumental in shaping this work.

I am grateful to my project supervisor at the National Institute of Technology Jalandhar, **Prof. V. Narasimha Nayak**, for his guidance and support during the academic progression of this work. I also thank **Prof. Awasthi K.** for facilitating the internship opportunity that enabled this collaboration.

I would like to acknowledge the guidance and technical mentorship provided by **Ramanathan C.** and **Venkatesh C. R.** at IIT Madras. Their inputs were crucial in understanding practical aspects of signal processing and implementation.

I also acknowledge the support and infrastructure provided by **Data Patterns (India) Pvt. Ltd., Chennai**, which contributed to the practical realization and validation of this work.

This work was carried out at the **Information Systems Laboratory (ISL)** and the **Intel Wireless Laboratory (IWL)** at IIT Madras, and I am thankful for the resources and research environment provided by both laboratories.

Finally, I would like to thank my parents, my sister, and my friends, especially Chhavi and Raj for their constant support, encouragement, and motivation throughout this journey, without whom none of this would have been possible.

Chinmay Deo

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Chapter 1

Introduction

Direction-of-arrival (DoA) estimation is essential for autonomous sensing and array-processing systems [1]. Uniform circular arrays (UCAs) offer full 360° azimuth coverage, avoiding the end-fire degradation of uniform linear arrays (ULAs). While eigenstructure based algorithms like MUSIC provide high-resolution DoA estimates on UCAs [2], coherent signals from multipath or repeater jamming cause rank-deficient source covariance. This severely degrades conventional subspace methods, complicating robust DoA estimation on compact arrays [3].

Mapping the UCA manifold to a virtual ULA via phase-mode transformation restores a Vandermonde-like structure. This yields significant advantages, including FFT-friendly weights, Root-MUSIC compatibility, and a virtual aperture that can exceed the physical array [4]. However, these techniques rely on clean subspace separation. Standard criteria like AIC and MDL often fail under signal coherence because they assume a positive-definite source covariance [5, 6, 7].

Spatial smoothing mitigates coherence by averaging overlapping subarrays to restore covariance rank [8]. While well-established for ULAs [9] and beamspace UCAs [10], recent adaptive MUSIC techniques typically focus on large arrays with tens to hundreds of elements and limited coherent sources [11, 12]. Reliable line-of-sight (LoS) DoA detection in non-minimum-phase channels remains underexplored for highly compact arrays.

This work addresses this gap by combining mode-space spatial smoothing with UCA-RB-MUSIC on a five-element UCA. We introduce a dynamic, noise-variance-aware eigenvalue thresholding scheme for robust subspace separation under nonstationary noise. Parametric analysis demonstrates near-Cramér–Rao bound (CRB) performance above 5 dB SNR.

Importantly, this framework does not attempt full spatial scene reconstruction. For a compact five-element array, fully resolving a congested multipath environment is physically unfeasible. Instead, our objective is strictly **LoS target isolation**. By sacrificing the resolution of secondary interferers, mode-space smoothing ensures the true LoS target remains the dominant, stable peak in the spatial spectrum even when degrees of freedom are saturated.

Chapter 2

Review of Literature

Direction-of-arrival (DoA) estimation has been extensively studied in array signal processing, with a wide range of techniques developed for different array geometries and propagation conditions. This section reviews the key contributions relevant to coherent source environments, compact arrays, and beamspace processing methods.

2.1 Classical DoA Estimation Methods

Eigenstructure-based techniques such as Multiple Signal Classification (MUSIC) have been widely adopted due to their high-resolution capabilities [2]. These methods rely on the separation of signal and noise subspaces through eigen-decomposition of the covariance matrix and assume that the source signals are uncorrelated. Under such conditions, MUSIC provides accurate estimation of source directions and has been extended to various array geometries, including uniform circular arrays (UCAs).

2.2 Coherent Source Problem

In practical environments, signals often arrive through multiple paths, resulting in coherent or highly correlated sources. This leads to rank deficiency in the covariance matrix, causing conventional subspace methods to fail [3]. The problem is particularly severe in non-minimum phase channels, where multipath components dominate the received signal and obscure the true line-of-sight (LOS) component.

2.3 Spatial Smoothing Techniques

Spatial smoothing was introduced to address the coherent source problem by dividing a uniform linear array (ULA) into overlapping subarrays and averaging their covariance matrices [8]. This approach decorrelates the sources and restores the rank of the covariance matrix. Further analysis of smoothing performance under correlated sources was provided in [9].

The extension of spatial smoothing to uniform circular arrays has been studied in [10], where beamspace transformations are used to enable smoothing in circular geometries. However, these approaches typically assume sufficiently large arrays and may not directly apply to compact or undersampled configurations.

2.4 Beamspace and Phase-Mode Processing

Phase-mode excitation provides a mechanism to transform the UCA manifold into a beamspace representation with a Vandermonde-like structure [13]. This transformation allows the application of techniques originally developed for ULAs. The eigenstructure-based formulation for UCAs and its beamspace interpretation have been well established in [2].

Mode-space spectral estimation techniques further exploit this transformation to perform DoA estimation in the modal domain [14]. These methods enable efficient processing and provide a framework for applying spatial smoothing in UCAs.

2.5 Model Order Selection

Accurate estimation of the number of sources is critical for subspace-based methods. Information-theoretic criteria such as Akaike Information Criterion (AIC) [5] and Minimum Description Length (MDL) [6, 7] are commonly used for this purpose. However, these methods assume uncorrelated sources and positive-definite covariance matrices, and their performance degrades significantly in the presence of coherent multipath.

2.6 Recent Advances and Practical Challenges

Recent work has explored low-complexity DoA estimation techniques for coherent sources [3] and adaptive beamforming methods for jamming mitigation [11]. Additionally, real-data validation of DoA estimation techniques has been investigated using alternative formulations such as propagator-based methods [12].

Despite these advancements, most existing studies consider large arrays with sufficient aperture or focus on resolving all sources in the scene. Limited attention has been given to highly compact and undersampled arrays, where spatial aliasing and reduced degrees of freedom impose fundamental constraints on resolvability.

2.7 Research Gap

From the above discussion, it is evident that while significant progress has been made in handling coherent sources and extending subspace methods to UCAs, challenges remain in compact array configurations. In such settings, complete reconstruction of all sources is often infeasible.

This work addresses this gap by focusing on reliable LOS detection rather than full scene reconstruction. By combining mode-space spatial smoothing, forward-backward averaging, and real beamspace MUSIC, the proposed approach enables robust LOS isolation in non-minimum phase channels using a five-element UCA. Additionally, the computational efficiency of the method is enhanced through FFT-based spectrum evaluation, and its effectiveness is validated using both simulated and real measurement data.

Chapter 3

Background

3.1 DoA Estimation Methods

DoA estimation techniques can be broadly categorized into beamforming, subspace-based, and parametric methods.

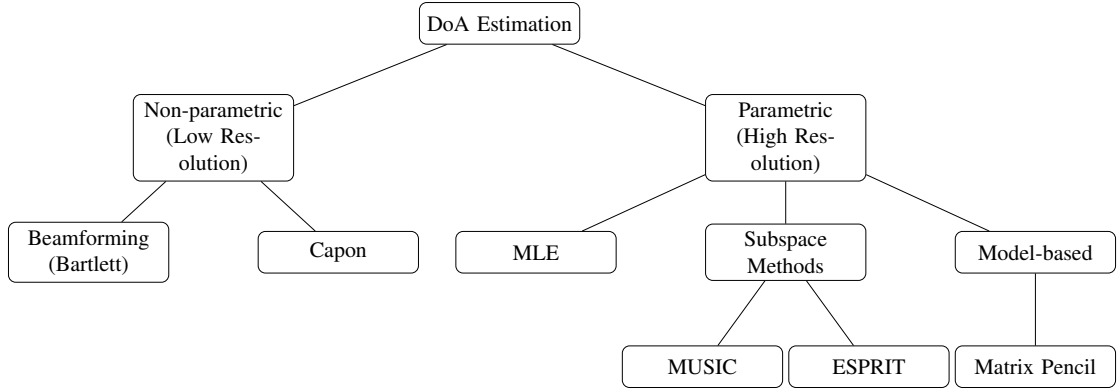


Figure 3.1: Classification of DoA estimation techniques

ESPRIT requires shift-invariant structures, Maximum Likelihood Estimation (MLE) is computationally expensive, and Matrix Pencil/Total Least Squares (TLS) methods are highly sensitive to noise and model mismatch in coherent environments. This work selects MUSIC for its high resolution, computational efficiency, and compatibility with beamspace transformations. MUSIC avoids ESPRIT's strict geometric constraints and MLE's expensive multi-dimensional optimizations. Although standard MUSIC degrades under signal coherence, spatial smoothing and forward-backward averaging effectively mitigate this limitation. Ultimately, MUSIC's adaptability to phase-mode processing provides an ideal foundation for Line-of-Sight (LOS) detection in compact UCA systems.

3.2 Statistical Properties of Signals and Subspaces

Subspace-based DoA algorithms rely on the statistical properties of the incoming signals [15]. The received signal vector is modeled as a Wide-Sense Stationary (WSS) random process $\mathbf{x} = [x(0), x(1), \dots, x(p)]^T$. Its second-order statistics are defined by

the autocorrelation matrix $\mathbf{R}_x$ and autocovariance matrix $\mathbf{C}_x$:

$$\mathbf{R}_x = \mathbb{E}[\mathbf{x}\mathbf{x}^H] \quad (3.1)$$

$$\mathbf{C}_x = \mathbf{R}_x - \mathbf{m}_x \mathbf{m}_x^H \quad (3.2)$$

where $\mathbf{m}_x$ is the mean vector. For zero-mean processes, $\mathbf{C}_x = \mathbf{R}_x$.

Subspace separation fundamentally relies on the **Eigenvalue Extremal Property**, which bounds the eigenvalues λ_i of the autocorrelation matrix by the extrema of the signal's Power Spectral Density (PSD), $P_x(e^{j\omega})$:

$$\min_{\omega} P_x(e^{j\omega}) \leq \lambda_i \leq \max_{\omega} P_x(e^{j\omega}) \quad (3.3)$$

This property guarantees that the covariance matrix can be cleanly partitioned into orthogonal signal and noise subspaces, provided their energies are well-separated.

3.2.1 Wold Decomposition and Rational Spectra

To model realistic narrowband signals rather than ideal pure tones, we apply the **Wold Decomposition theorem** [15]. This theorem dictates that any WSS random process comprises a predictable (deterministic) component and a regular (stochastic) component:

$$x(n) = x_r(n) + x_p(n) \quad (3.4)$$

Consequently, the Innovations Representation states that any regular process can be generated by filtering white noise $v(n)$ (with variance σ_v^2) through a causal, stable Linear Shift-Invariant (LSI) system $H(z)$.

If $H(z)$ is a rational transfer function, the output follows an Autoregressive Moving Average (ARMA) process. By configuring the poles of an Autoregressive (AR) filter, white noise is shaped into a narrowband process with a defined center frequency and finite bandwidth. The **Yule-Walker equations** relate the AR filter coefficients a_i to the signal's autocorrelation sequence $r_x(k)$:

$$r_x(k) + \sum_{i=1}^p a_i r_x(k-i) = \sigma_v^2 |b(0)|^2 \delta(k) \quad (3.5)$$

This statistical framework underpins the realistic signal generation models utilized throughout this work's methodology.

Chapter 4

Proposed Methodology and Theoretical Framework

This section establishes the signal model, provides a brief recap of the beamspace transform theory and spatial smoothing extended with forward-backward averaging (FBA), and then proposes an inequality for appropriate parameter selection. The parameters of interest are the noise variance threshold multiplier and the subarray count.

4.1 Signal Model

An incoming signal is modeled as a **uniform plane harmonic wave**:

$$s_n(t) = A \cos \left(2\pi f_c t + K_0 \hat{r} \cdot \vec{P}_n + \alpha \right) \quad (4.1)$$

where $\vec{P}_n$ is the position vector of the n^{th} element, $K_0 = \frac{2\pi}{\lambda}$ is the wave number, and $\hat{r}$ is the unit vector in the direction of arrival. In spherical coordinates, $\hat{r}$ is the vector of direction cosines.

Using complex notation:

$$s_n(t) = A e^{j\alpha} \cdot e^{jK_0 \hat{r} \cdot \vec{P}_n} \quad (4.2)$$

$$x_n(t) = \tilde{g}_0(t) \cdot e^{jK_0 \hat{r} \cdot \vec{P}_n} \quad (4.3)$$

where $\tilde{g}_0(t) = A e^{j\alpha}$ is the complex envelope at the origin.

4.2 Steering Vector

The **manifold** or **steering vector** is a length- N vector representing the response of an N -element array to a signal arriving from direction $\hat{r}$. It encodes the relative phase shifts across the array elements due to wave propagation:

$$\underline{\mathbf{a}}(\hat{r}) = \begin{bmatrix} e^{jK_0 \hat{r} \cdot \vec{P}_0} \\ e^{jK_0 \hat{r} \cdot \vec{P}_1} \\ \vdots \\ e^{jK_0 \hat{r} \cdot \vec{P}_{N-1}} \end{bmatrix} \quad (4.4)$$

Array Pattern is the sum of the responses described by the steering vector and is treated as the array output.

For a Uniform Circular Array (UCA) of radius r in the $\theta = 90^\circ$ plane (XY-plane), the position vector of the n^{th} element is:

$$\vec{P}_n = \begin{bmatrix} r \cos \gamma_n \\ r \sin \gamma_n \\ 0 \end{bmatrix}, \quad \text{where } \gamma_n = \frac{2\pi(n-1)}{N} \quad (4.5)$$

Using complex envelope notation, the UCA steering vector becomes:

$$\underline{\mathbf{a}}(\theta, \phi) = \begin{bmatrix} e^{j\zeta \cos(\phi-\gamma_0)} \\ e^{j\zeta \cos(\phi-\gamma_1)} \\ \vdots \\ e^{j\zeta \cos(\phi-\gamma_{N-1})} \end{bmatrix}, \quad \text{where } \zeta = K_0 r \sin \theta \quad (4.6)$$

The array data model is:

$$\mathbf{X}(t) = \mathbf{A} \cdot \mathbf{S}(t) + \mathbf{n}(t) \quad (4.7)$$

where $\mathbf{S}(t)$ is the signal vector at the reference point, and $\mathbf{n}(t)$ is the noise.

4.3 Phase-Mode Excitation

Instead of weighting individual UCA elements arbitrarily, the weights are defined as a function of each element's angular position:

$$w(\gamma_n) = e^{jm\gamma_n}, \quad \text{where } \gamma_n = \frac{2\pi(n-1)}{N} \quad (4.8)$$

m is the **mode**, an integer defining the spatial “harmonic”. Now, instead of synthesizing patterns element by element, we can directly excite the entire array into a mode [13]. These **spatial harmonics**, just like in Fourier series, can form arbitrary patterns by superposition:

$$w(\gamma) = \sum_{m=-M}^M c_m e^{jm\gamma} \quad (4.9)$$

Applying mode- m weights to the UCA element outputs and summing gives the array output:

$$y_m(t) = \sum_{n=0}^{N-1} x_n(t) \cdot e^{jm\gamma_n} \quad (4.10)$$

This operation transforms the array from **element space** (individual sensors) to **beam space** (modal domain). A maximum of M modes can be obtained, defined by $M = \lfloor K_0 r \rfloor$, where r is the radius of the circular array.

The far-field array pattern for mode m ($N > 2M$, which is the equivalent of the Nyquist criterion) becomes:

$$A_m(\phi) = j^m \cdot J_m(\zeta) \cdot e^{jm\phi} \quad (4.11)$$

where $J_m(\zeta)$ is the Bessel function of the first kind (order m) and $\zeta = k_0 r \sin \theta$, and ϕ is the azimuth angle. The exponential term $e^{jm\phi}$ introduces a **Vandermonde-like structure** as observed in the uniform linear array (ULA) manifolds. Thus, we essentially transform the UCA into a virtual ULA. We will also show that our study holds valid for heavily aliased compact arrays, such as a five-element UCA.

4.4 Beamforming Matrices

The beamformer matrix $\mathbf{F}$ is designed such that $\mathbf{F}\mathbf{F}^H = \mathbf{I}$, so as to prevent coloring of white noise. The matrix of modal weights is defined as:

$$\mathbf{V} = \sqrt{N} [\mathbf{w}_{-M} \cdots \mathbf{w}_0 \cdots \mathbf{w}_M] \quad (4.12)$$

$$\mathbf{w}_m = \frac{1}{N} [e^{jm\gamma_0}, \dots, e^{jm\gamma_{N-1}}]^T \quad (4.13)$$

Its Hermitian $\mathbf{V}^H \in \mathbb{C}^{(2M+1) \times N}$ has each row as a beamformer for a spatial mode. The **even** (complex) beamformer matrix is defined as:

$$\mathbf{F}_e^H = \mathbf{C}_e \mathbf{V}^H, \quad \mathbf{C}_e = \text{diag}(j^{-M}, \dots, j^{-1}, 1, j^1, \dots, j^M) \quad (4.14)$$

To obtain a real-valued beamspace representation, we define:

$$\mathbf{F}_r^H = \mathbf{W}^H \mathbf{F}_e^H \quad (4.15)$$

$$\mathbf{W}^H = \frac{1}{\sqrt{M'}} \begin{bmatrix} \mathbf{v}(\alpha_{-M})^H \\ \vdots \\ \mathbf{v}(\alpha_0)^H \\ \vdots \\ \mathbf{v}(\alpha_M)^H \end{bmatrix}, \quad \alpha_i = \frac{2\pi i}{M'} \quad (i = -M, \dots, M) \quad (4.16)$$

The resulting real-valued beamspace manifold is:

$$\mathbf{a}_r(\theta, \phi) = \mathbf{F}_r^H \mathbf{a}(\theta, \phi) = \sqrt{N} \cdot \mathbf{W}^H \mathbf{J}(\zeta) \mathbf{v}(\phi) \quad (4.17)$$

And the array output can be expressed as:

$$\mathbf{a}_r(\theta, \phi) = \begin{bmatrix} f(\zeta, \phi - \alpha_{-M}) \\ \vdots \\ f(\zeta, \phi - \alpha_0) \\ \vdots \\ f(\zeta, \phi - \alpha_M) \end{bmatrix} \quad (4.18)$$

Where:

$$f(\zeta, \phi) = \sqrt{\frac{N}{M'}} \left[J_0(\zeta) + 2 \sum_{m=1}^M J_m(\zeta) \cos(m\phi) \right] \in \mathbb{R} \quad (4.19)$$

The detailed theory behind mode-space excitation for uniform circular arrays and its interpretation has been established in [2] and [13].

4.5 Mode-space Spatial Smoothing

We split the M' dimension virtual ULA into S ($S = M' - M'' + 1$) smaller M'' dimension ($M'' \leq M' - D + 1$) sub-arrays to perform spatial smoothing in mode space [14]. D denotes the number of sources. To further enhance de-correlation, forward-backward

averaging can be used. We have chosen to retain the Bessel function based tapering unlike previous works.

$$R_{fb} = \frac{1}{2S} \sum_{l=1}^S Z_l^T [R + \bar{I}(R)^* \bar{I}] Z_l \quad (4.20)$$

where

$$Z_l = [e_l, e_{l+1}, \dots, e_{l+M''-1}] \quad (4.21)$$

$$M'' = 2M - S + 2 \quad (4.22)$$

with e_l denoting the l^{th} column of the identity matrix of size $2M + 1$. S is the number of sub-arrays, $(\cdot)^*$ denotes complex conjugate, and $\bar{I}$ is the exchange matrix.

The extent of de-correlation post-smoothing is governed by a Dirichlet kernel factor [10] given by the effective correlation coefficient:

$$|\rho_f| = \left| \rho \frac{\sin(S(\theta_i - \theta_j)/2) \cos \psi}{S \sin((\theta_i - \theta_j)/2)} \right| \quad (4.23)$$

Here, ψ is the phase angle of the complex correlation coefficient ($\rho = |\rho|e^{j\psi}$). ρ_f depends *only* on the angular separation $(\theta_i - \theta_j)$, not on individual directions, and is completely independent of inter-element spacing of array elements. In a ULA, correlation depends on both inter-element spacing and individual directions.

4.6 Dynamic Noise Variance Thresholding and Subspace Separation

Successful Direction-of-Arrival (DoA) estimation requires reliable separation of the smallest signal eigenvalue from the noise subspace. After spatial smoothing, the raw pairwise signal correlations are scaled by the Dirichlet kernel (Section II-B). For strictly coherent multipath sources, where the pre-smoothing correlation magnitude is $|\rho| = 1$, the effective $D \times D$ post-smoothing correlation matrix C is given by

$$C_{ij} = \begin{cases} 1, & i = j \\ \rho_f(\theta_i, \theta_j), & i \neq j \end{cases} \quad (4.24)$$

where $\rho_f(\theta_i, \theta_j)$ denotes the spatial smoothing decorrelation between the i -th and j -th sources for S subarrays. The smoothed signal covariance matrix is

$$\bar{R}_s = P^{1/2} C P^{1/2} \quad (4.25)$$

with source powers $P = \text{diag}(P_1, \dots, P_D)$.

To approximate the smallest signal eigenvalue $\lambda_D(\bar{R}_s)$, we use the matrix trace. Since the coherent signal subspace is highly ill-conditioned, λ_D dominates the inverse sum:

$$\text{Tr}(\bar{R}_s^{-1}) = \sum_{i=1}^D \frac{1}{\lambda_i} \approx \frac{1}{\lambda_D(\bar{R}_s)} \quad (4.26)$$

$$\lambda_D(\bar{R}_s) \approx \left\{ \text{Tr}(P^{-1/2} C^{-1} P^{-1/2}) \right\}^{-1} = \left\{ \sum_{i=1}^D \frac{[C^{-1}]_{ii}}{P_i} \right\}^{-1} \quad (4.27)$$

This approximation provides an interpretable link between decorrelation strength and the minimum recoverable signal energy and is most accurate when the sources are well-separated.

Relating the total data covariance matrix R_x to the smoothed source covariance gives $\lambda_D(R_x) = \lambda_D(A_s \overline{R_s} A_s^H) + \sigma_n^2$. Since the non-zero eigenvalues of XY equal those of YX , and assuming nearly orthogonal beamspace steering vectors ($A_s^H A_s \approx M'' I$), the array gain $M'' = M' - S + 1$ factors out:

$$\lambda_D(R_x) \approx (M' - S + 1) \left\{ \sum_{i=1}^D \frac{P_i}{[C^{-1}]_{ii}} \right\} + \sigma_n^2 \quad (4.28)$$

For successful partitioning, this smallest signal eigenvalue must exceed the noise threshold $K\sigma_n^2$, where K is a variance multiplier. This yields the baseline inequality for source enumeration:

$$(M' - S + 1) \left\{ \sum_{i=1}^D \frac{P_i}{[C^{-1}]_{ii}} \right\} \geq (K - 1)\sigma_n^2 \quad (4.29)$$

Equation (4.29) shows that S and K must be jointly selected so that the decorrelated signal energy exceeds the threshold margin.

Applying this inequality requires a reliable estimate of the true noise variance, σ_n^2 . To make the criterion practical, we use an empirical dynamic thresholding heuristic that adapts to trial-specific leakage energy. The trial-specific thermal noise variance, $\hat{\sigma}_n^2$, is estimated by averaging the smallest p eigenvalues:

$$\hat{\sigma}_n^2 = \frac{1}{p} \sum_{i=M''-p+1}^{M''} \lambda_i \quad (4.30)$$

The dynamic signal subspace boundary is then defined as

$$\gamma = K \cdot \hat{\sigma}_n^2 \quad (4.31)$$

Eigenvalues satisfying $\lambda_i > \gamma$ are classified as belonging to the restored signal subspace.

4.7 Realistic Modeling and Generation of Narrowband Signals

Conventionally, incoming signals in Direction-of-Arrival (DoA) estimation are modeled as ideal pure tones. In discrete time, the ideal complex exponential is represented as $s(n) = Ae^{j\omega_0 n}$, which corresponds to a single frequency component with an infinitely narrow spectrum. While mathematically convenient, pure tones are idealized representations that fail to capture practical hardware imperfections and finite coherence times.

To better represent real-world signal variations, our validation models incoming targets as narrowband stochastic processes exhibiting a finite bandwidth around the center frequency ω_0 . These processes are generated by filtering white noise $v(n)$ through linear shift-invariant autoregressive (AR) systems.

4.7.1 Complex Exponential Representation (AR(1) Model)

For a complex narrowband process characterized by a single spectral peak at ω_0 , a single pole representation is sufficient. This is achieved using a first-order Autoregressive, or AR(1), model. The system function for an AR(1) process is given by:

$$H(z) = \frac{1}{1 - az^{-1}} \quad (4.32)$$

which yields a Power Spectral Density (PSD) of:

$$P_x(e^{j\omega}) = \frac{\sigma_v^2}{|1 - ae^{-j\omega}|^2} \quad (4.33)$$

where σ_v^2 is the variance of the driving white noise. By placing the pole a near the unit circle at the desired normalized center frequency, a sharp but finite-width spectral peak is generated.

4.7.2 Real Sinusoid Representation (AR(2) Model)

If the signal is alternatively modeled as a real sinusoid, such as $s(n) = A \cos(\omega_0 n)$, its spectrum contains peaks at both $\pm\omega_0$. This requires a conjugate pole pair to model the two spectral peaks, necessitating a second-order AR(2) band-pass process:

$$H(z) = \frac{1}{(1 - az^{-1})(1 - a^*z^{-1})}, \quad \text{where } a = re^{j\omega_0} \quad (4.34)$$

The resulting PSD is:

$$P_x(e^{j\omega}) = \frac{\sigma_v^2}{|1 - 2r \cos(\omega_0)e^{-j\omega} + r^2e^{-j2\omega}|^2} \quad (4.35)$$

where r controls the bandwidth of the signal. Values of r approaching 1 (e.g., $r = 0.9$) narrow the bandwidth, approximating a pure tone, while smaller values widen it.

4.7.3 Simulation Implementation

For our algorithmic validation, the signals were generated using the complex AR(1) model. The parameters were configured to emulate a narrowband process with a center frequency of $f_0 = 702$ kHz and a sampling frequency of $f_s = 1.6$ MHz.

As illustrated in Figure 4.1, moving from an ideal pure tone to the AR(1) narrowband stochastic model replaces the infinitely narrow Dirac delta peak with a realistic, finite-bandwidth spectral profile. This ensures our Monte Carlo simulations test the mode-space spatial smoothing algorithm under conditions that closely mirror the realities of actual RF environments.

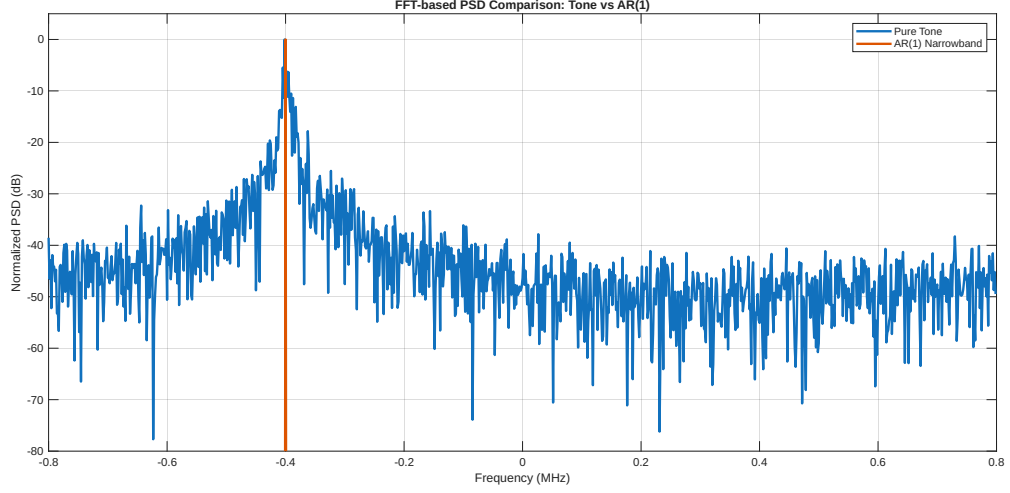


Figure 4.1: Comparison of the Power Spectral Density (PSD) between an ideal pure tone and the realistic AR(1) narrowband stochastic model.

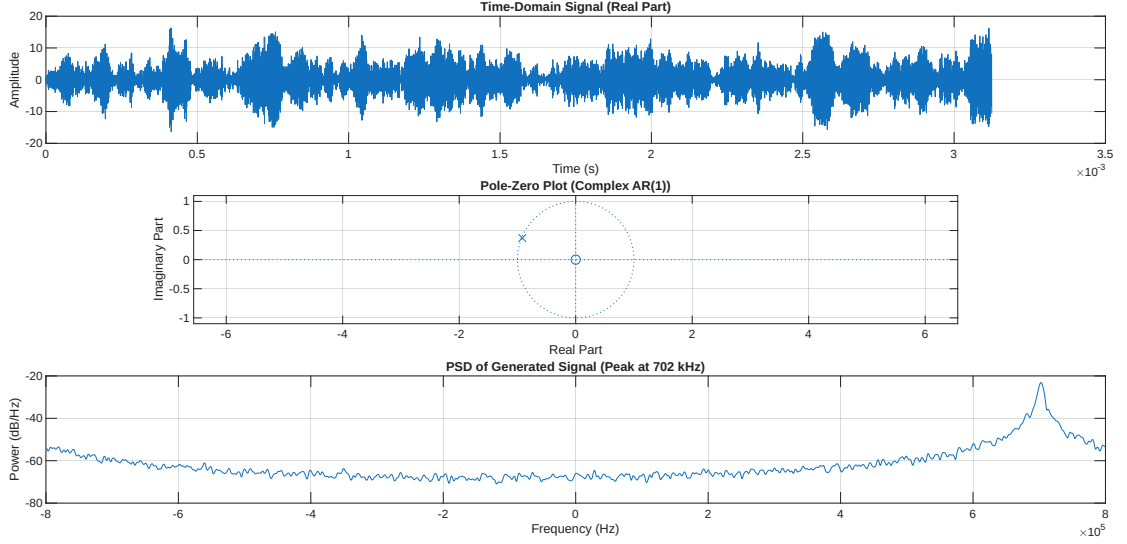


Figure 4.2: Observed spectral peak of the generated complex AR(1) process at $f_0 = 702$ kHz.

4.8 FFT Interpretation of RB-MUSIC

Conventional MUSIC requires an exhaustive 2D grid search over elevation (θ) and azimuth (ϕ), evaluating the projection of the steering vector onto the noise subspace at every coordinate. This $O(N^2)$ approach is computationally prohibitive.

By leveraging the Vandermonde-like structure restored by the phase-mode transformation, the noise subspace projection in beamspace can be expressed as:

$$P(\theta, \phi) \propto \frac{1}{\mathbf{V}^H(\phi) \mathbf{J}(\zeta) (\mathbf{W} \mathbf{V}_n \mathbf{V}_n^T \mathbf{W}^H) \mathbf{J}(\zeta) \mathbf{V}_r(\phi)} \quad (4.36)$$

where $\zeta = k_0 r \sin \theta$, and $\mathbf{V}_n$ is the estimated noise subspace.

To decouple the elevation and azimuth dependencies, we define an elevation-dependent core matrix $\mathbf{Q}_\zeta$:

$$\mathbf{Q}_\zeta = \mathbf{J}_\zeta \cdot \mathbf{W} \cdot \mathbf{V}_n \mathbf{V}_n^T \cdot \mathbf{W}^H \cdot \mathbf{J}_\zeta \quad (4.37)$$

Because $\mathbf{Q}_\zeta$ is entirely independent of ϕ , the denominator—termed the **Null Spectrum**—can be formulated as a polynomial evaluated on the unit circle:

$$P_{\text{Null}}(\phi) = \sum_{\ell=-(M'-1)}^{M'-1} a_\zeta(\ell) \cdot e^{j\ell\phi} \quad (4.38)$$

where the coefficients $a_\zeta(\ell)$ represent the diagonal sums of $\mathbf{Q}_\zeta$:

$$a_\zeta(\ell) = \sum_{\substack{i,j \\ j=i+\ell}} [\mathbf{Q}_\zeta]_{i,j} \quad (4.39)$$

Exploiting the Hermitian symmetry of $\mathbf{Q}_\zeta$, we define a one-sided symmetric coefficient sequence:

$$a'_\zeta = \{a_\zeta(0), 2a_\zeta(1), \dots, 2a_\zeta(M'-1)\} \quad (4.40)$$

Substituting this sequence into the null spectrum equation yields:

$$P_{\text{Null}}(\phi) = \sum_{\ell=0}^{M'-1} a'_\zeta(\ell) \cdot e^{-j\ell\phi} \quad (4.41)$$

This expression is mathematically identical to the Discrete Fourier Transform (DFT). Consequently, the entire azimuthal search for a given elevation reduces to a single FFT operation:

$$P_{\text{Null}} = \text{FFT}(a'_\zeta)[l], \quad \phi_l = \frac{2\pi l}{L_{fft}}, \quad l = 0, 1, \dots, L_{fft} - 1 \quad (4.42)$$

Computational Advantage: The matrix $\mathbf{Q}_\zeta$ is constructed only once per elevation step. A 0° to 89° elevation scan at a 1° resolution requires exactly 90 FFT operations. By zero-padding the input sequence to an FFT length of $N_{fft} = 360$, the output bins map directly to a 1° azimuthal grid (0° to 359°). This $O(N \log N)$ acceleration delivers a near $10\times$ speedup over the baseline spatial search.

4.9 Cramér–Rao Bound for Azimuth Estimation

To benchmark the proposed estimator against a theoretical lower bound, the deterministic Cramér–Rao Bound (CRB) for azimuth estimation is evaluated [16]. The CRB represents the minimum achievable variance of any unbiased estimator under the assumed signal model and therefore serves as an important reference for assessing the asymptotic efficiency of the proposed UCA-RB-MUSIC framework.

Since the performance evaluation in this work focuses primarily on azimuth estimation accuracy, the CRB is derived only for the azimuth parameter ϕ , while the elevation angle is fixed at $\theta = 90^\circ$.

Assuming D uncorrelated narrowband sources, the source covariance matrix is modeled as

$$\mathbf{R}_s = \text{diag}(P_1, P_2, \dots, P_D) \quad (4.43)$$

where P_i denotes the power of the i^{th} source. Under the deterministic signal model, the azimuth Fisher Information Matrix (FIM) is given by [17]

$$\mathbf{J}_\phi = \frac{2T}{\sigma_n^2} \Re \{ (\mathbf{D}_\phi^H \mathbf{P}_\mathbf{A}^\perp \mathbf{D}_\phi) \circ \mathbf{R}_s^T \} \quad (4.44)$$

where T denotes the number of temporal snapshots, σ_n^2 is the additive white Gaussian noise variance, and $\circ$ denotes the Hadamard product. The matrix

$$\mathbf{P}_\mathbf{A}^\perp = \mathbf{I} - \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \quad (4.45)$$

represents the orthogonal projection operator onto the noise subspace, while

$$\mathbf{D}_\phi = \left[\frac{\partial \mathbf{a}(\phi_1)}{\partial \phi_1}, \dots, \frac{\partial \mathbf{a}(\phi_D)}{\partial \phi_D} \right] \quad (4.46)$$

contains the steering-vector gradients with respect to azimuth.

For an N -element Uniform Circular Array (UCA), the steering response of the n^{th} sensor is expressed as

$$a_n(\theta, \phi) = \exp(j K_0 r \sin \theta \cos(\phi - \gamma_n)) \quad (4.47)$$

where $K_0 = \frac{2\pi}{\lambda}$ denotes the wave number, r is the array radius, and

$$\gamma_n = \frac{2\pi(n-1)}{N} \quad (4.48)$$

represents the angular position of the n^{th} array element.

Differentiating the steering response with respect to azimuth yields

$$\frac{\partial a_n(\theta, \phi)}{\partial \phi} = -j K_0 r \sin \theta \sin(\phi - \gamma_n) a_n(\theta, \phi) \quad (4.49)$$

The deterministic CRB matrix is therefore obtained as

$$\text{CRB}(\phi) = \mathbf{J}_\phi^{-1} \quad (4.50)$$

Finally, the theoretical lower bound on the Root Mean Square Error (RMSE) for the k^{th} source azimuth estimate is computed as

$$\text{RMSE}_{\text{CRB},k} = \frac{180}{\pi} \sqrt{[\mathbf{J}_\phi^{-1}]_{kk}} \quad (4.51)$$

This CRB formulation is used in Chapter 5 to benchmark the proposed mode-space smoothed UCA-RB-MUSIC estimator against the theoretical estimation limit under varying Signal-to-Noise Ratio (SNR) conditions. The complete derivation is elaborated in Appendix B.

Chapter 5

Results and Analysis

This chapter evaluates the performance of the proposed UCA-RB-MUSIC framework through simulation and real-data experiments. The evaluation begins by demonstrating the limitations of conventional model-order selection methods and subspace separation techniques in coherent multipath environments. The effectiveness of the proposed mode-space spatial smoothing and dynamic thresholding framework is then examined across specular-dominant and fully scattered propagation conditions. Comparative analyses of smoothed and unsmoothed UCA-RB-MUSIC are presented through spatial spectra, eigenvalue distributions, and RMSE performance relative to the theoretical Cramér–Rao Bound (CRB). Finally, the operational limits of the framework under coherent interference are investigated, followed by validation using real field measurements obtained with a five-element Uniform Circular Array.

5.1 Experimental Setup

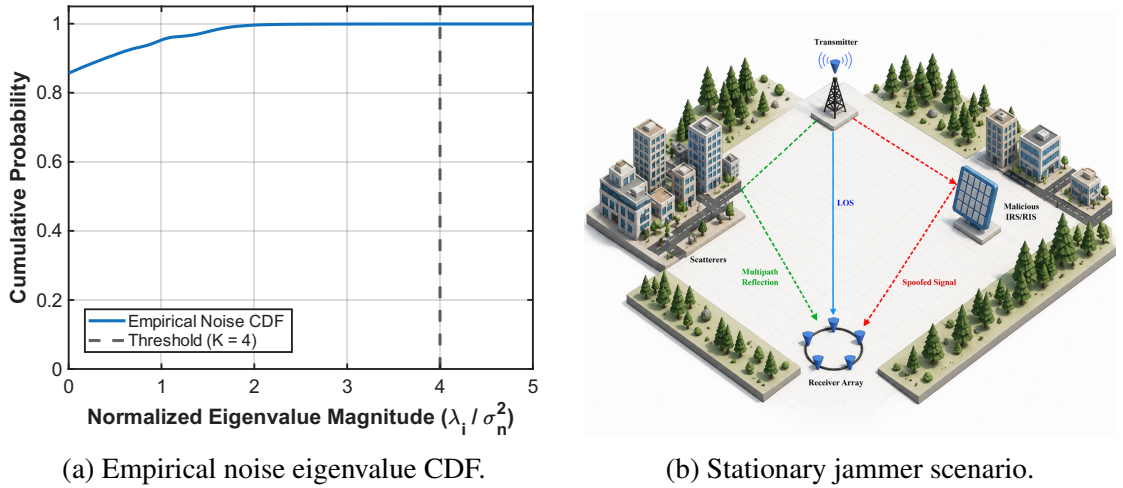


Figure 5.1: Experimental environment and noise characteristics used for evaluation.

The proposed mode-space smoothing UCA-RB-MUSIC framework was evaluated using Monte Carlo simulations (16 trials, $T = 128$ snapshots) on a compact five-element UCA with radius $r = 0.5095$ m. A narrowband AR(1) signal with carrier frequency $f_c = 702$ MHz, sampling frequency $f_s = 1.6$ MHz, and correlation coefficient $\rho =$

0.995 was used, yielding an effective bandwidth of approximately 2.5 kHz. Coherent multipath was generated using delayed replicas of the common waveform with short, medium, and long delay spreads corresponding to delay sets $[0, 1, 2, 3]$, $[0, 10, 20, 30]$, and $[0, 100, 200, 300]$ samples, respectively, with $[0, 1, 2, 3]$ used unless otherwise specified. Relative source powers were set to $[-3, 0, -6, -12]$ dB, and the ground-truth configuration consisted of four coplanar emitters with $\theta = 90^\circ$ and $\phi = [55^\circ, 65^\circ, 85^\circ, 100^\circ]$. Performance was evaluated at an operational SNR of 9 dB under the stationary jammer scenario shown in Fig. 5.1(b) across both specular-dominant (open-terrain combat zone) and fully scattered (urban combat zone) propagation environments. Following Pillai [18], forward-backward averaging (FBA) used at least $S = 2$ subarrays to restore covariance matrix rank for four coherent sources. The noise variance $\hat{\sigma}_n^2$ was estimated from the smallest eigenvalue ($p = 1$) [15], while the threshold multiplier $K = 4$ was selected from the empirical noise eigenvalue distribution in Fig. 5.1(a).

5.2 Limitations of Model Order Selection in Coherent Multipath

Under the open-terrain combat scenario, both the Akaike Information Criterion (AIC) and Minimum Description Length (MDL) criterion produced incorrect model-order estimates. The dominant MUSIC spectrum peak aligned with the jammer, resulting in incorrect source identification and degraded localization performance (Table 5.1, Fig. 5.2). These results highlight the failure of conventional information-theoretic model-order selection methods in coherent multipath and interference-dominated environments with non-minimum phase channels.

Table 5.1: Comparison of Dominant Peak Estimates Under Different Model-Order Selection Methods

Case	Model Order Selection Criterion	Estimated Peaks (ϕ°, θ°)	Power (dB)
Unsmoothed	Manual	(65, 82)	0.00
		(273, 29)	-2.12
		(333, 89)	-2.41
		(176, 43)	-2.82
Smoothed	AIC	(65, 90)	0.00
		(57, 85)	-0.15
		(101, 75)	-4.19
	MDL	(65, 90)	0.00
		(55, 89)	-0.08
		(101, 89)	-4.58
	Our Method	(55, 90)	0.00
		(62, 89)	-0.71
		(269, 26)	-1.98
		(273, 26)	-4.27

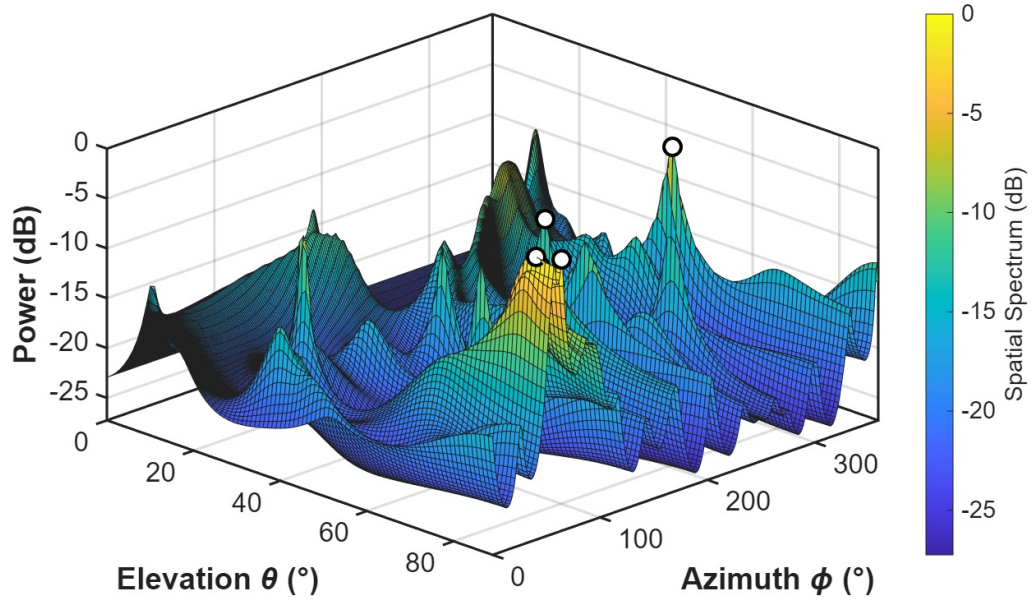


Figure 5.2: Spatial spectrum using the AIC estimate.

5.3 Localization Performance Across Multipath Regimes

In Table 5.2, correctly identified first peaks are shown in blue, while incorrect first peaks are shown in red. The results show that the first detected peak remains stable across short and medium delay spreads in the specular-dominant environment, correctly identifying the LOS direction. However, under long delay spread, the dominant peak shifts to $(65^\circ, 90^\circ)$, indicating degraded localization performance due to increased temporal decorrelation. In the fully scattered environment, the algorithm correctly identifies the LOS direction only under short delay spread, while medium and long delay regimes produce dominant peaks near the jammer direction $(65^\circ, 90^\circ)$, resulting in incorrect source identification.

Table 5.2: First-peak estimates under both channel conditions

Scenario	Channel Condition	Delay	Peak (ϕ, θ)
Open terrain	Specular-dominant	Short	$(55, 90)$
Open terrain	Specular-dominant	Medium	$(55, 90)$
Open terrain	Specular-dominant	Long	$(65, 90)$
Urban	Fully scattered	Short	$(55, 90)$
Urban	Fully scattered	Medium	$(65, 90)$
Urban	Fully scattered	Long	$(65, 90)$

5.4 Comparative Analysis: Smoothed vs. Unsmoothed UCA-RB-MUSIC

This section examines the effect of mode-space spatial smoothing through three complementary views: the MUSIC pseudo-spectra in Figs. 5.3 and 5.4, the covariance eigenvalue structure in Fig. 5.5, and the RMSE behavior against the theoretical CRB in Fig. 5.6. The results show that smoothing restores subspace separability, reduces spurious responses, and improves tracking accuracy under coherent multipath.

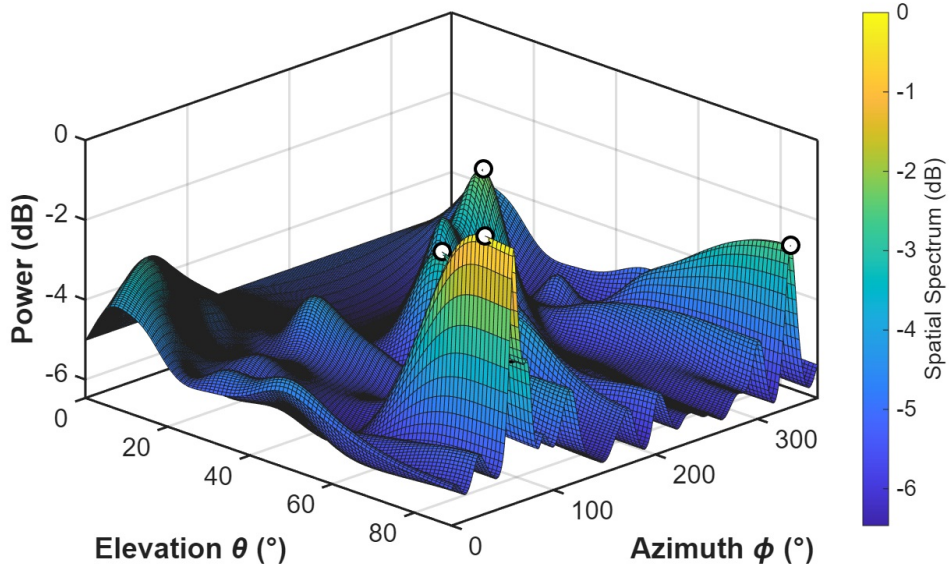


Figure 5.3: MUSIC spectrum without spatial smoothing.

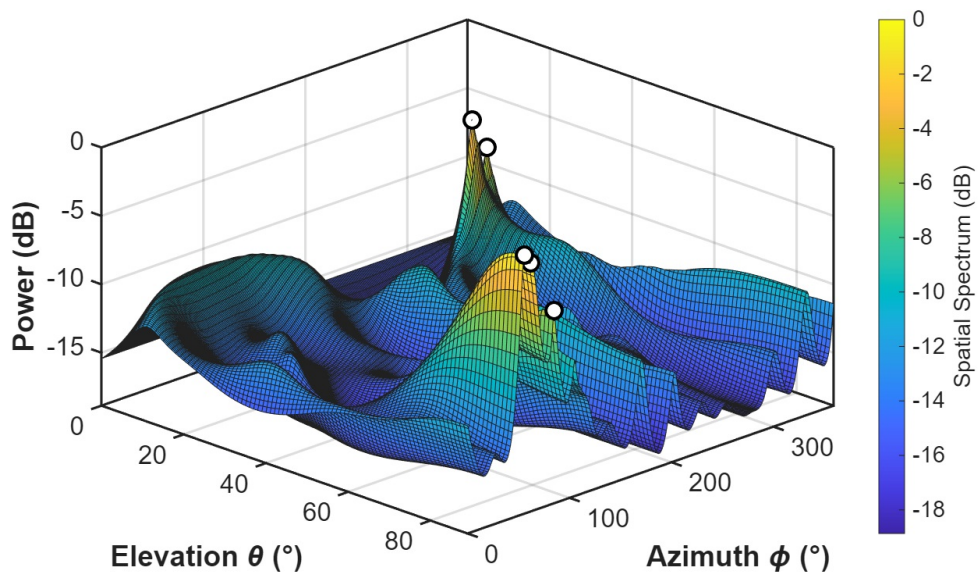


Figure 5.4: MUSIC spectrum after mode-space spatial smoothing.

Without smoothing, the covariance matrix remains rank-deficient, with only two dominant eigenvalues and the remaining eigenvalues clustered near the noise floor. As shown in Fig. 5.3, this produces multiple spurious peaks in the MUSIC pseudo-spectrum, with the dominant response aligned with the jammer rather than the true line-of-sight (LOS) source.

After applying spatial smoothing, four dominant eigenvalues become clearly separable, enabling stable model-order estimation through the proposed thresholding scheme. The corresponding pseudo-spectrum in Fig. 5.4 shows improved peak separation, correct elevation recovery, and fewer spurious responses. The dominant peak estimates are summarized in Table 5.1. Although a secondary spurious peak appears at -0.71 dB, temporal averaging of successive spatial spectra can mitigate peak swapping and stabilize long-term LOS tracking.

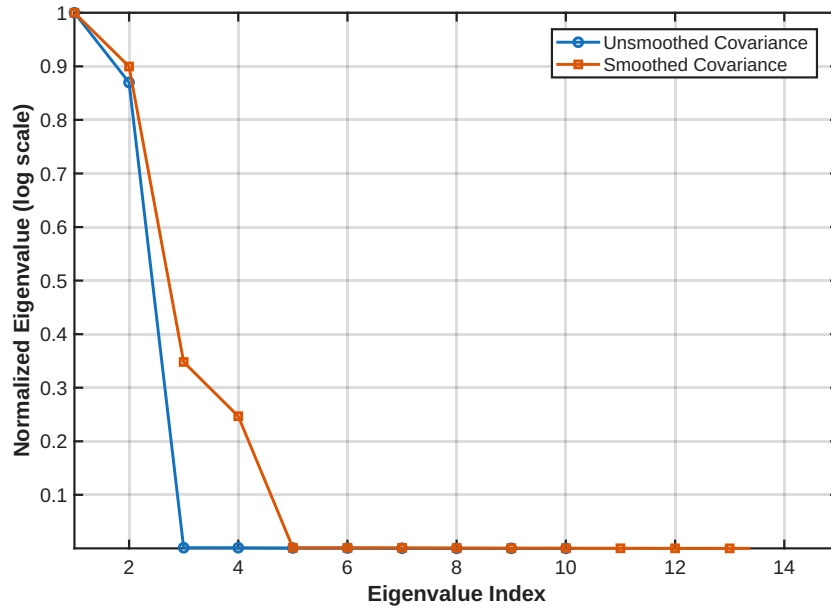


Figure 5.5: Eigenvalue distribution before and after spatial smoothing.

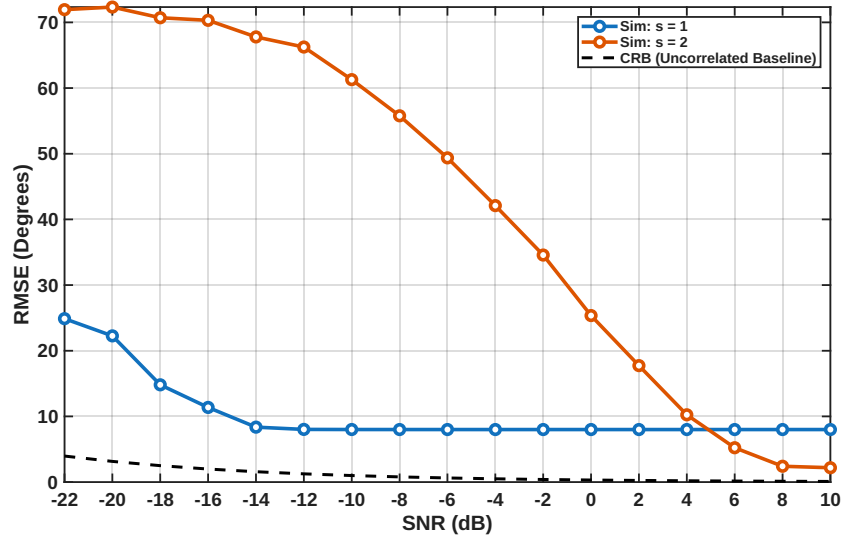


Figure 5.6: RMSE comparison between standard UCA-RB-MUSIC ($s = 1$) and the proposed mode-space smoothed variant ($s = 2$) against the theoretical CRB.

Figure 5.5 confirms the underlying reason for the spectral improvement. The unsmoothed case retains only two dominant eigenvalues, which is insufficient for reliable separation of four coherent sources. After smoothing, the signal subspace becomes well conditioned and the dominant eigenvalues are restored, allowing the proposed thresholding rule to identify the signal and noise subspaces more robustly.

Figure 5.6 compares RMSE versus operating SNR for the smoothed and unsmoothed variants. To emphasize coherent multipath effects, the third and fourth delayed components were assigned relative powers of -12 dB. Each RMSE point was computed from 400 Monte Carlo trials using the first detected peak relative to the ground-truth LOS direction. The reported operating SNR includes coherent integration across $T = 128$ snapshots, corresponding to an approximate processing gain of $10 \log_{10}(128) \approx 21$ dB relative to the per-snapshot input SNR.

At low SNR, both methods exhibit large errors due to noise-dominated covariance estimation. As SNR increases, the unsmoothed variant saturates near an RMSE of approximately 8° because of covariance rank deficiency and jammer-dominated peak selection, whereas the proposed smoothed variant continues improving and approaches the deterministic azimuth CRB for the LOS source at operating SNRs above approximately 5 dB.

5.5 Operational Boundaries and Jammer Tolerance

To quantify the operational limits of the proposed framework under adversarial conditions, a Jammer-to-Signal Ratio (JSR) stress test was conducted. The LOS source power was fixed at 0 dB while a coherent repeater jammer was swept from 0 to +30 dB.

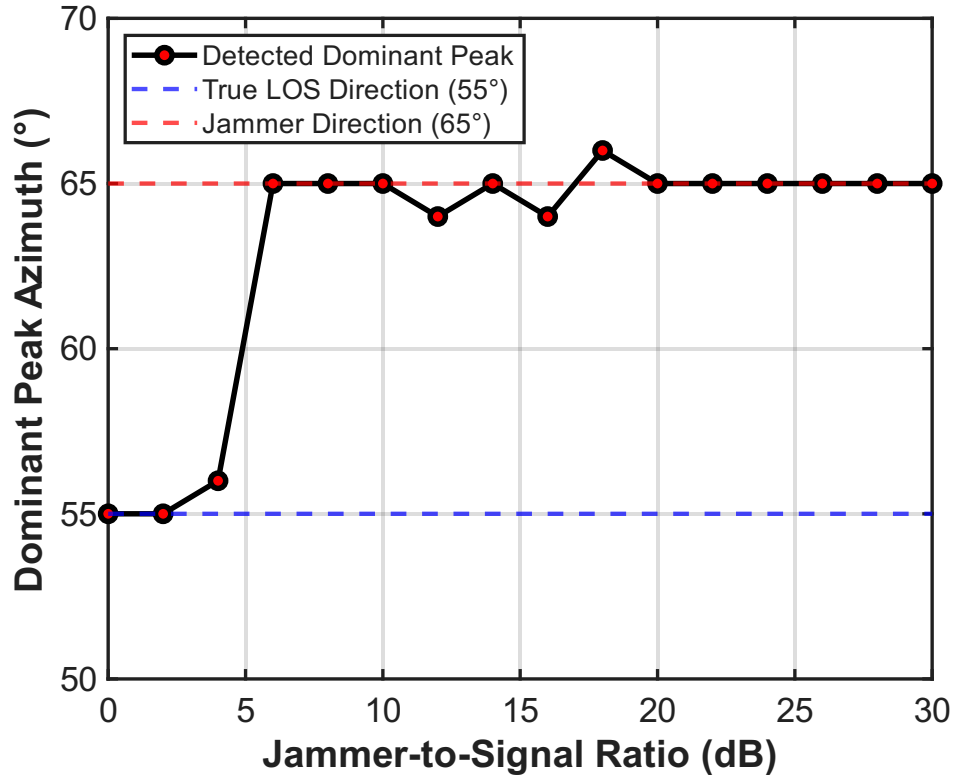


Figure 5.7: Dominant detected azimuth as a function of jammer-to-signal ratio (JSR).

Fig. 5.7 shows the dominant detected azimuth as a function of JSR. The proposed framework preserves the LOS direction as the dominant peak up to approximately +4 dB JSR. Beyond this point, the dominant peak shifts to the jammer direction (65°), indicating breakdown of LOS isolation.

The observed +4 dB tolerance therefore defines an approximate operational boundary for the proposed compact-array architecture. Stronger coherent jammers would require either increased array aperture or additional temporal interference suppression prior to spatial processing.

5.6 Real-Time Validation

To validate the proposed algorithm's performance against realistic hardware impairments and physical multipath scattering in a controlled outdoor environment, real-time field tests were conducted in collaboration with our industry partner, Data Patterns.

5.6.1 Experimental Setup and Calibration Pipeline

The field deployment utilized a 5-element Uniform Circular Array (UCA) as the receiver. Both the transmitting and receiving antennas were mounted at an operating height of 12 m, with a Line-of-Sight (LoS) separation distance of 130 m. The physical setup established a ground truth target at an azimuth of 320° and an elevation of 90° (horizon). The receiver hardware configuration is illustrated in Figure 5.8.



Figure 5.8: The 5-element UCA field deployment utilized for real-time data acquisition.

The recorded raw dataset comprised 2048 IQ snapshots. To rigorously evaluate the algorithm's temporal consistency, these data were partitioned into 16 independent Monte Carlo trials, processing 128 snapshots per trial.

Before spatial processing, an antenna calibration pipeline was applied to compensate for inherent phase mismatches across the UCA elements. Calibration involved applying a cumulative phase rotation to each element relative to Antenna 1, given by:

$$X_{cal,i} = X_{raw,i} \cdot e^{j\phi_i}$$

where $X_{cal,i}$ and $X_{raw,i}$ represent the calibrated and raw baseband signals respectively, and ϕ_i is the measured phase offset for the i -th element.

5.6.2 Five-element UCA: Field Data Validation

The performance of the proposed Mode-Space Smoothing algorithm was benchmarked against the conventional (Old) method and Explicit Forward-Backward Averaging (FBA). The arrays were tested across a wide frequency sweep ranging from 252 MHz to 812 MHz. A detection was considered successful (highlighted in green) if the estimated azimuth fell within a $\pm 5^\circ$ window around the 320° ground truth (i.e., $[315^\circ, 325^\circ]$).

As detailed in Table 5.3 (values represent the first detected peak in the averaged spectra), the standard method fails completely at higher frequencies due to spatial aliasing and hardware-induced phase noise. Explicit FBA manages to stabilize the estimates at lower frequencies but breaks down entirely above 600 MHz. In contrast, the proposed Mode-Space Smoothing technique demonstrates superior robustness, successfully recovering the true target azimuth across the entire upper frequency band despite the hardware impairments.

Freq (MHz)	Standard (Az, El, P)	With FBA (Az, El, P)	Mode-Space Smoothing (Az, El, P)
252	322°, 90°, -5.7	320°, 90°, 0.0	340°, 73°, 0.0
412	152°, 19°, -2.6	320°, 90°, 0.0	320°, 86°, 0.0
622	52°, 30°, -4.7	147°, 36°, 0.0	317°, 77°, 0.0
647	102°, 17°, 17.4	149°, 33°, 0.0	316°, 72°, 0.0
702	150°, 23°, -2.3	319°, 90°, 0.0	319°, 89°, 0.0
712	148°, 23°, 0.4	322°, 90°, 0.0	322°, 89°, 0.0
707	-	149°, 23°, 3.0	320°, 89°, 0.0
812	161°, 14°, 2.3	163°, 12°, 0.0	320°, 89°, 0.0

Table 5.3: 5-element UCA DoA detection performance using field data.

Chapter 6

Conclusion

This work investigated the limits of a compact five-element Uniform Circular Array (UCA) for line-of-sight (LOS) recovery in coherent and adversarial RF environments with non-minimum phase channels. By combining mode-space transformation, spatial smoothing, and forward-backward averaging, the proposed UCA-RB-MUSIC framework restores covariance rank under coherent multipath conditions and enables stable subspace separation through dynamic noise-variance thresholding. Across LOS-dominant and fully scattered propagation scenarios, the method consistently improves localization stability, approaches the deterministic Cramér–Rao Bound (CRB) at appreciable operational SNR, and maintains robust LOS isolation under moderate coherent interference. These results demonstrate the practical viability of compact UCAs for sensing, electronic support, and passive radar applications in interference-dominated environments.

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Appendices

Appendix A

Statistical Analysis of the Signal Eigenvalue Inequality

This appendix provides an analytical derivation of a sufficient condition governing the smallest signal-space eigenvalue λ_D of the spatially smoothed covariance matrix. The result establishes a conservative bound for reliable LOS detection under coherent multipath and interference.

The analysis considers spatial smoothing without forward-backward averaging. Since FBA introduces additional decorrelation and noise averaging, the derived condition should be interpreted as conservative relative to the implemented system.

Let the spatially smoothed signal covariance matrix be expressed as

$$\overline{\mathbf{R}}_s = \mathbf{P}^{1/2} \mathbf{C} \mathbf{P}^{1/2} \quad (\text{A.1})$$

where $\mathbf{P} = \text{diag}(P_1, \dots, P_D)$ contains the source powers, and $\mathbf{C}$ is a normalized correlation matrix with unit diagonal and coherence-dependent off-diagonal entries induced by multipath and spatial smoothing.

Taking the inverse,

$$\overline{\mathbf{R}}_s^{-1} = \mathbf{P}^{-1/2} \mathbf{C}^{-1} \mathbf{P}^{-1/2} \quad (\text{A.2})$$

and therefore

$$\text{Tr}(\overline{\mathbf{R}}_s^{-1}) = \sum_{i=1}^D \frac{[C^{-1}]_{ii}}{P_i}. \quad (\text{A.3})$$

For a positive definite matrix with eigenvalues $\{\lambda_i\}$,

$$\text{Tr}(\overline{\mathbf{R}}_s^{-1}) = \sum_{i=1}^D \frac{1}{\lambda_i}. \quad (\text{A.4})$$

In strongly coherent scenarios, the covariance matrix becomes ill-conditioned, and the smallest eigenvalue dominates the inverse sum. Under this approximation,

$$\lambda_D(\overline{\mathbf{R}}_s) \approx \left(\sum_{i=1}^D \frac{[C^{-1}]_{ii}}{P_i} \right)^{-1}. \quad (\text{A.5})$$

To interpret the diagonal entries of $\mathbf{C}^{-1}$, we note that $\mathbf{C}$ is a correlation matrix with unit diagonal. Using the Schur complement, the diagonal elements of its inverse are

$$[C^{-1}]_{ii} = \frac{1}{1 - R_i^2} \quad (\text{A.6})$$

where R_i^2 is the multiple correlation coefficient of the i^{th} source with respect to all other sources.

Substituting this expression yields

$$\lambda_D(\bar{\mathbf{R}}_s) \approx \left(\sum_{i=1}^D \frac{1}{P_i(1 - R_i^2)} \right)^{-1}. \quad (\text{A.7})$$

This expression shows that the smallest eigenvalue is jointly governed by source power and coherence. High coherence ($R_i^2 \rightarrow 1$) collapses the denominator, reducing the effective rank, while low source power further suppresses the eigenvalue.

For interpretability, a conservative approximation can be obtained by considering the dominant contribution from the weakest and most correlated source. In this regime,

$$\lambda_D(\bar{\mathbf{R}}_s) \sim P_{\min}(1 - R_{\max}^2), \quad (\text{A.8})$$

which captures the limiting behavior as coherence approaches unity. This approximation is not exact, but provides a useful lower-bound scaling relation.

The total covariance matrix is given by

$$\mathbf{R}_x = \mathbf{A}_s \bar{\mathbf{R}}_s \mathbf{A}_s^H + \sigma_n^2 \mathbf{I}. \quad (\text{A.9})$$

Using the property that the non-zero eigenvalues of $\mathbf{XY}$ and $\mathbf{YX}$ are identical,

$$\lambda_D(\mathbf{A}_s \bar{\mathbf{R}}_s \mathbf{A}_s^H) = \lambda_D(\mathbf{A}_s^H \mathbf{A}_s \bar{\mathbf{R}}_s). \quad (\text{A.10})$$

Under beam-space transformation, the steering matrix satisfies the approximate orthogonality condition

$$\mathbf{A}_s^H \mathbf{A}_s \approx (M' - S + 1) \mathbf{I}, \quad (\text{A.11})$$

where $(M' - S + 1)$ represents the effective aperture after mode-space transformation and spatial smoothing.

Thus,

$$\lambda_D(\mathbf{R}_x) \approx (M' - S + 1) \lambda_D(\bar{\mathbf{R}}_s) + \sigma_n^2. \quad (\text{A.12})$$

A sufficient condition for reliable subspace separation is that the smallest signal eigenvalue exceeds the noise-scaled threshold:

$$(M' - S + 1) \lambda_D(\bar{\mathbf{R}}_s) + \sigma_n^2 \geq K \sigma_n^2. \quad (\text{A.13})$$

Substituting the conservative approximation yields

$$(M' - S + 1) P_{\min}(1 - R_{\max}^2) \gtrsim (K - 1) \sigma_n^2. \quad (\text{A.14})$$

This bound characterizes the interplay between array aperture, coherence, source power, and detection threshold. Increasing aperture improves separability, while high coherence and low signal power reduce the effective signal subspace dimension, making LOS detection more challenging.

Appendix B

Cramér–Rao Bound for DoA Azimuth Estimation

The Cramér–Rao Bound (CRB) provides the fundamental lower bound on the variance of any unbiased estimator. In array signal processing, the deterministic or conditional CRB treats the unknown source waveforms as deterministic sequences and therefore gives a benchmark conditioned on the realized source signals and array geometry. This appendix derives the deterministic CRB for joint azimuth estimation using a Uniform Circular Array (UCA) under the independent-source reference model used throughout this work.

Consider an N -element sensor array receiving narrowband signals from D sources over T snapshots. The observed baseband data vector at snapshot t is modeled as

$$\mathbf{X}(t) = \mathbf{A}(\boldsymbol{\phi})\mathbf{S}(t) + \mathbf{N}(t), \quad t = 1, 2, \dots, T \quad (\text{B.1})$$

where $\boldsymbol{\phi} = [\phi_1, \phi_2, \dots, \phi_D]^T \in \mathbb{R}^{D \times 1}$ denotes the unknown azimuth angles, $\mathbf{A}(\boldsymbol{\phi}) = [\mathbf{a}(\phi_1), \mathbf{a}(\phi_2), \dots, \mathbf{a}(\phi_D)] \in \mathbb{C}^{N \times D}$ is the steering matrix, $\mathbf{S}(t) \in \mathbb{C}^{D \times 1}$ contains the deterministic source waveforms, and $\mathbf{N}(t)$ is zero-mean circularly symmetric spatially white Gaussian noise with covariance

$$\mathbb{E}[\mathbf{N}(t)\mathbf{N}^H(s)] = \sigma_n^2 \mathbf{I} \delta_{t,s}.$$

Conditioned on these parameters, the likelihood of the complete data matrix $\mathbf{X} = [\mathbf{X}(1), \dots, \mathbf{X}(T)]$ is

$$p(\mathbf{X}; \boldsymbol{\phi}, \mathbf{S}, \sigma_n^2) = \frac{1}{(\pi\sigma_n^2)^{NT}} \exp \left(-\frac{1}{\sigma_n^2} \sum_{t=1}^T \|\mathbf{X}(t) - \mathbf{A}(\boldsymbol{\phi})\mathbf{S}(t)\|^2 \right). \quad (\text{B.2})$$

Taking the natural logarithm gives

$$\mathcal{L}(\boldsymbol{\phi}, \mathbf{S}, \sigma_n^2) = -NT \ln(\pi\sigma_n^2) - \frac{1}{\sigma_n^2} \sum_{t=1}^T \|\mathbf{X}(t) - \mathbf{A}(\boldsymbol{\phi})\mathbf{S}(t)\|^2. \quad (\text{B.3})$$

The Fisher Information Matrix (FIM) quantifies the sensitivity of the likelihood to perturbations in the unknown parameters. Eliminating the deterministic signal parameters $\mathbf{S}(t)$ via partial differentiation, Stoica and Nehorai [16, 17] showed that the concentrated FIM for the spatial parameter vector $\boldsymbol{\phi}$ can be written using the orthogonal projector onto the array noise subspace:

$$\mathbf{P}_A^\perp = \mathbf{I} - \mathbf{A}(\boldsymbol{\phi}) (\mathbf{A}^H(\boldsymbol{\phi})\mathbf{A}(\boldsymbol{\phi}))^{-1} \mathbf{A}^H(\boldsymbol{\phi}) \quad (\text{B.4})$$

and the derivative matrix

$$\mathbf{D}_\phi = \left[\frac{\partial \mathbf{a}(\phi_1)}{\partial \phi_1}, \frac{\partial \mathbf{a}(\phi_2)}{\partial \phi_2}, \dots, \frac{\partial \mathbf{a}(\phi_D)}{\partial \phi_D} \right] \in \mathbb{C}^{N \times D}. \quad (\text{B.5})$$

The sample source covariance across T snapshots is

$$\mathbf{R}_s = \frac{1}{T} \sum_{t=1}^T \mathbf{S}(t) \mathbf{S}^H(t). \quad (\text{B.6})$$

Under the independent-source reference model, this reduces to

$$\mathbf{R}_s = \text{diag}(P_1, P_2, \dots, P_D), \quad (\text{B.7})$$

where P_k is the average power of the k^{th} source.

Using the concentrated asymptotic formulation of Stoica and Nehorai [16, 17], the FIM for joint azimuth estimation is

$$\mathbf{J}_\phi = \frac{2T}{\sigma_n^2} \Re \{ (\mathbf{D}_\phi^H \mathbf{P}_\mathbf{A}^\perp \mathbf{D}_\phi) \circ \mathbf{R}_s^T \} \quad (\text{B.8})$$

where $\Re\{\cdot\}$ denotes the real-part operator and $\circ$ represents the Hadamard product.

For an N -element UCA of radius r in the XY-plane, the angular position of the n^{th} sensor is

$$\gamma_n = \frac{2\pi(n-1)}{N}, \quad n = 1, 2, \dots, N. \quad (\text{B.9})$$

The steering response of the n^{th} element to a source arriving from azimuth ϕ_k is

$$a_n(\phi_k) = \exp(jK_0 r \cos(\phi_k - \gamma_n)), \quad (\text{B.10})$$

where $K_0 = \frac{2\pi}{\lambda}$ is the free-space wavenumber. Differentiating with respect to ϕ_k gives

$$\frac{\partial a_n(\phi_k)}{\partial \phi_k} = -jK_0 r \sin(\phi_k - \gamma_n) a_n(\phi_k). \quad (\text{B.11})$$

Stacking the derivatives yields

$$\frac{\partial \mathbf{a}(\phi_k)}{\partial \phi_k} = \begin{bmatrix} -jK_0 r \sin(\phi_k - \gamma_1) a_1(\phi_k) \\ -jK_0 r \sin(\phi_k - \gamma_2) a_2(\phi_k) \\ \vdots \\ -jK_0 r \sin(\phi_k - \gamma_N) a_N(\phi_k) \end{bmatrix}. \quad (\text{B.12})$$

By the Cramér–Rao inequality, the covariance of any unbiased estimator $\hat{\phi}$ is lower-bounded by the inverse FIM:

$$\text{CRB}(\phi) = \mathbf{J}_\phi^{-1}. \quad (\text{B.13})$$

Thus, the minimum achievable variance for each azimuth estimate is

$$\text{var}(\hat{\phi}_k) \geq [\mathbf{J}_\phi^{-1}]_{kk}, \quad k = 1, 2, \dots, D. \quad (\text{B.14})$$

For the target line-of-sight source, the RMSE lower bound in degrees is

$$\text{RMSE}_{\text{CRB}}(\phi_{\text{LOS}}) = \frac{180}{\pi} \sqrt{[\mathbf{J}_\phi^{-1}]_{\text{LOS}, \text{LOS}}}. \quad (\text{B.15})$$

Because spatial smoothing sacrifices effective array aperture in order to restore covariance rank, comparing the measured RMSE of the smoothed estimator against this independent-source CRB provides a direct measure of the asymptotic efficiency and residual decorrelation penalties of the compact five-element sub-array architecture.